Evidence of Evolution

Background

When Charles Darwin first proposed the idea that all new species descend from an ancestor, he performed an exhaustive amount of research to provide as much evidence as possible. Today, the major pieces of evidence for this theory can be broken down into the fossil record, embryology, comparative anatomy and molecular biology.

Fossils

The fossil record for the ancestors of the modern horse is one of the most complete records.

Cut out the pictures of the ancestors of the horse, their leg bones and skulls on the back page. Glue them in order of their evolutionary progression from the ancestor, Hyracotherium, to the modern horse, Equus.

For each stage, describe the change in

- Body shape
- Skull shape
- Leg shape

Hyracotherium

Body Shape:

_____

_____

Skull Shape:

_____

_____

Leg Shape: _____ _____

Mesohippus

Body Shape:

_____

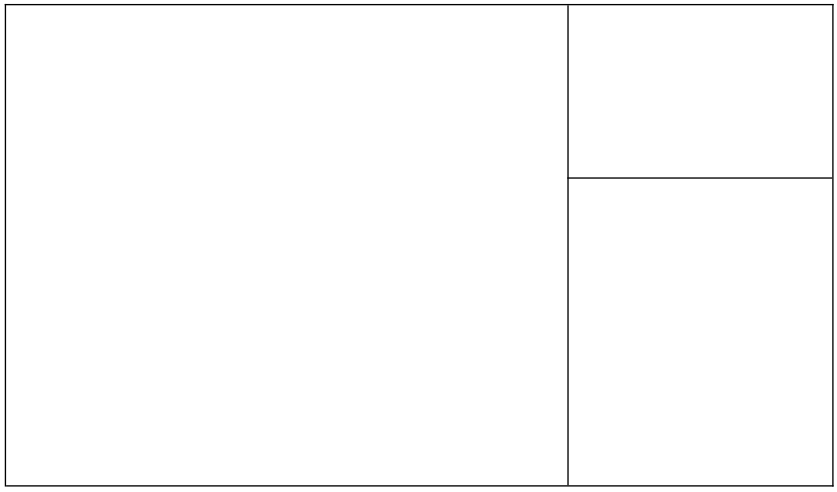
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Skull Shape:

_____

_____

Leg Shape: _____ _____



Merychippus

Body Shape:

_____

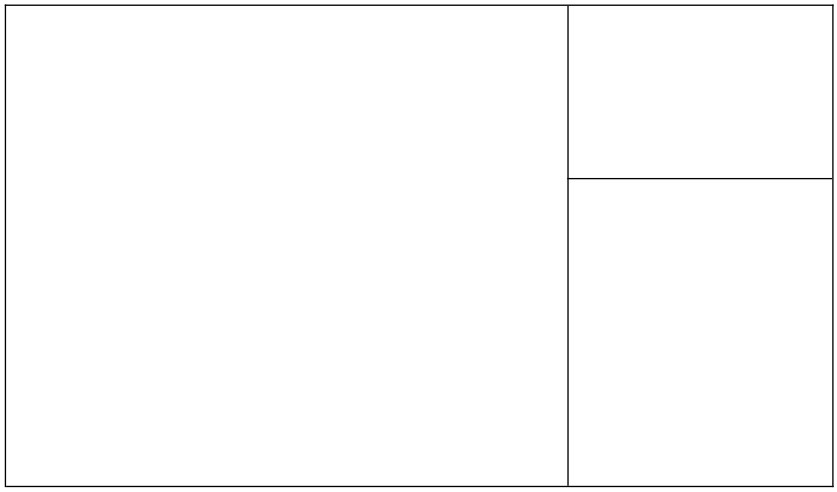
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Skull Shape:

_____

_____

Leg Shape: _____ _____



Pliohippus

Body Shape:

_____

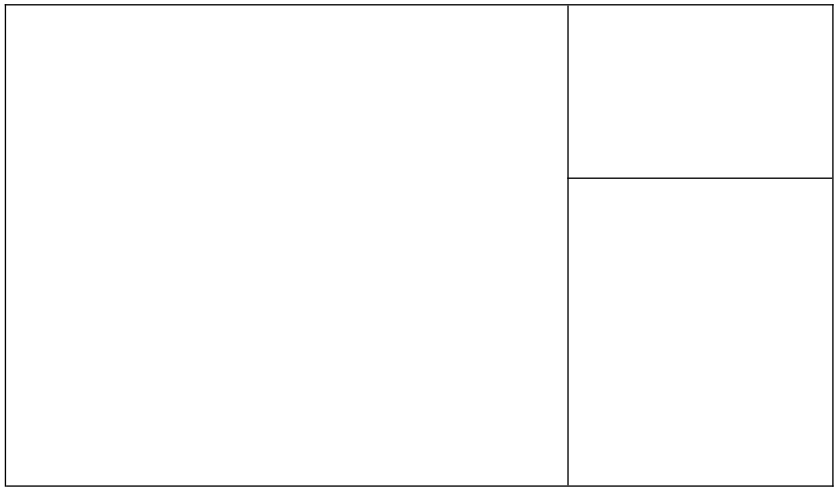
_____

Skull Shape:

_____

_____

Leg Shape: _____ _____



Equus

Body Shape:

_____

_____

Skull Shape:

_____

_____

Leg Shape: _____ _____

What is the biggest change in skull anatomy that occurred from Hyracotherium to Equus?

What was the biggest change in leg anatomy that occurred from Hyracotherium to Equus?

Embryology

Organisms that are closely related may also have physical similarities before they are even born. Look at the 6 different embryos below.

Hypothesise which embryo is from each of the following organisms and give a reason for your choice.

Species	Embryo	Reason
Human		
Chicken		
Rabbit		
Tortoise		
Salamander		
Fish		

These are older, more developed embryos from the same organisms.

Hypothesise which embryo is from each of the following organisms and give a reason for your choice.

Species	Embryo	Reason
Human		
Chicken		

Rabbit		
Tortoise		
Salamander		
Fish		

These are the embryos at their most advanced stage, shortly before birth.

Describe how the embryos have changed for each of these organisms from their earliest stage to the latest stage.

Species	**Anatomical changes From Early to Late Stages**
Human	
Chicken	
Rabbit	
Tortoise	
Salamander	
Fish	

Look again at the embryos in their earliest stages. What physical similarities exist between each of the embryos?

Does this suggest an evolutionary relationship? Can these embryos be used as evidence of a common ancestor between each of these six organisms?

Comparative Anatomy

Shown below are images of the skeletal structure of the front limbs of 6 animals: human, crocodile, whale, cat, bird and bat. Each animal has a similar set of bones.

Use the labelled human arm to colour code the different bones in each limb. Complete the key below.

Humerous	☐	Carpals	☐
Ulna	☐	Metacarpals	☐
Radius	☐	Phlanges	☐

For each animal, indicate what type of movement each limb is responsible for.

Animal	Primary Functions
Human	Using tools, picking up and holding objects
Whale	
Crocodile	
Cat	
Bird	
Bat	

Compare the skeletal structure of each limb to the human arm. Relate the differences you see in *form* to the differences in *function*.

Animal	Comparison to human arm in form	Comparison to human arm in function
Whale	Much shorter and thicker humerous, radius and ulna. Much longer metacarpals. Thumb has been shortened to a stub.	Whale fin needs to be longer to help movement through water. Thumbs not necessary as fins are not used for grasping.
Crocodile		
Cat		
Bird		

Bat		

Compare the anatomy of the butterfly and bird wing below.

What is the function of each of these structures?

How are they different in form?

Compare the body structure of the cave fish and the minnow below.

What is the most obvious difference between the body structures of the two fish?

Assume the fish came from the same ancestor. Why might the cave fish have evolved without eyes?

What kind of sensory adaptation would you hypothesise the cave fish to have to allow it to navigate in a cave and catch its food?

Molecular Biology

Cytochrome c is a protein found in mitochondria. It is used in the study of evolutionary relationships because most animals have this protein. Cytochrome c is made of 104 amino acids joined together. Below is a list of some of the amino acids in part of a Cytochrome c protein molecule for 9 different animals. Not all amino acids are shown.

For each non-human animal, take a highlighter and mark any amino acids that are different to the human sequence. When you finish record how many differences you found in the table on the next page.

	Human	Chicken	Horse	Tuna	Frog	Shark	Turtle	Monkey	Rabbit
42	Q	Q	Q	Q	Q	Q	Q	Q	Q
43	A	A	A	A	A	A	A	A	A
44	P	E	P	E	A	Q	E	P	V
46	Y	F	F	Y	F	F	F	Y	F
47	S	S	T	S	S	S	S	S	S
49	T	T	T	T	T	T	T	T	T
50	A	D	D	D	D	D	E	A	D
53	K	K	K	K	K	K	K	K	K
54	N	N	N	S	N	S	N	N	N
55	K	K	K	K	K	K	K	K	K
56	G	G	G	G	G	G	G	G	G
57	I	I	I	I	I	I	I	I	I
58	I	T	T	V	T	T	T	T	T
60	G	G	K	N	G	Q	G	G	G
61	E	E	E	N	E	Q	E	E	E
62	D	D	E	D	D	E	E	D	D
63	T	T	T	T	T	T	T	T	T
64	L	L	L	L	L	L	L	L	L
65	M	M	M	M	M	R	M	M	M
66	E	E	E	E	E	I	E	E	E
100	K	D	K	S	S	K	D	K	K
101	A	A	A	A	A	T	A	A	A
102	T	T	T	T	C	A	T	T	T
103	N	S	N	S	A	A	S	N	N
104	E	K	E	-	K	S	K	E	E

Animal	No. of Amino Acids Different to Humans	Animal	No. of Amino Acids Different to Humans
Horse		Shark	
Chicken		Turtle	
Tuna		Monkey	
Frog		Rabbit	

Based on the Cytochrome c data, which animal is most closely related to humans?

Do any of the animals have the same number of differences from human Cytochrome c? In situations like this, how would you decide what is more closely related to humans?

Conclusion

Charles Darwin published his book *On the Origin of Species* in 1859. Of the different types of evidence that you have looked at, which do you think he relied upon the most? Give reasons for your answer.

Given the amount of research and evidence available, why is it still classified as a theory?